

Supplementary Information for “Resistance-Resilient Antimalarial Leads from African Natural Products: MD Validation Against Resistance Mutants”

July 25, 2026

Table 1: Docking protocol validation: enrichment metrics and re-docking accuracy for the four wild-type targets. Values reproduce the corrected benchmarks from the parent study (V2-corrected grids; AutoDock Vina + DiffDock consensus). MMV Malaria Box values are consensus-score ROC-AUCs; DEKOIS 2.0 validation used AutoDock Vina only. BEDROC values use $\alpha = 20$ (MMV). Re-docking RMSD is the heavy-atom RMSD between the top-scoring docked pose and the co-crystallised ligand; MTX redocking failed (RMSD ≈ 30 Å).

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDROC	RMSD (Å)
PfDHFR (7F3Y)	Vina redock	–	–	–	–	<2.0
	MMV consensus (Vina+DiffDock)	0.924	–	2.57	1.309	–
	DEKOIS (Vina only)	0.509	–	0.51	–	–
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	–	1.11	9.434	–
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810	–
PfClpP (4GM2)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810	–

Note.—Redocking against V2-corrected grids gave 100% success for the five alignable co-crystallised ligands. MTX (PfDHFR co-crystallised ligand) could not be redocked and is reported as N/A. AutoDock Vina + Glna consensus scoring was used in the present validation study; the enrichment metrics above are taken from the parent study (Vina + DiffDock) to ensure consistency with the corrected P1 data.

Table 2: MM-GBSA binding free energies (ΔG_{bind} , kcal mol^{−1}) for the polypharmacological candidates against wild-type targets. Mean $\pm$ standard deviation from 101 snapshots. Due to GAFF2/CHARMM36 parameter conversion limitations (bond energy overflow errors), mutant calculations were computationally unfeasible; docking-based RRS is used as the primary validation metric.

Ligand	Target	ΔG_{bind}	Std. dev.	Status / RRS Class
164	PfClpP-WT	−8.35 [‡]	2.54 [‡]	Unbound [‡] (ligand dissociated)
201	PfDHFR-WT	−24.74 [‡]	4.63 [‡]	Unbound [‡] (Class A*)
214	PfCRT-WT	−18.25	0.40	Validated (gmx_MMPBSA) (Class B)
438	PfATP4-WT	–	–	Excluded [§] (Class A)
87	PfDHFR-WT	–	–	Not simulated (Class C)

[‡]Binds to allosteric site (NADPH-domain interface), not catalytic site.

[§]Excluded due to two-chain topology incompatibility with CHARMM36-to-AMBER conversion.

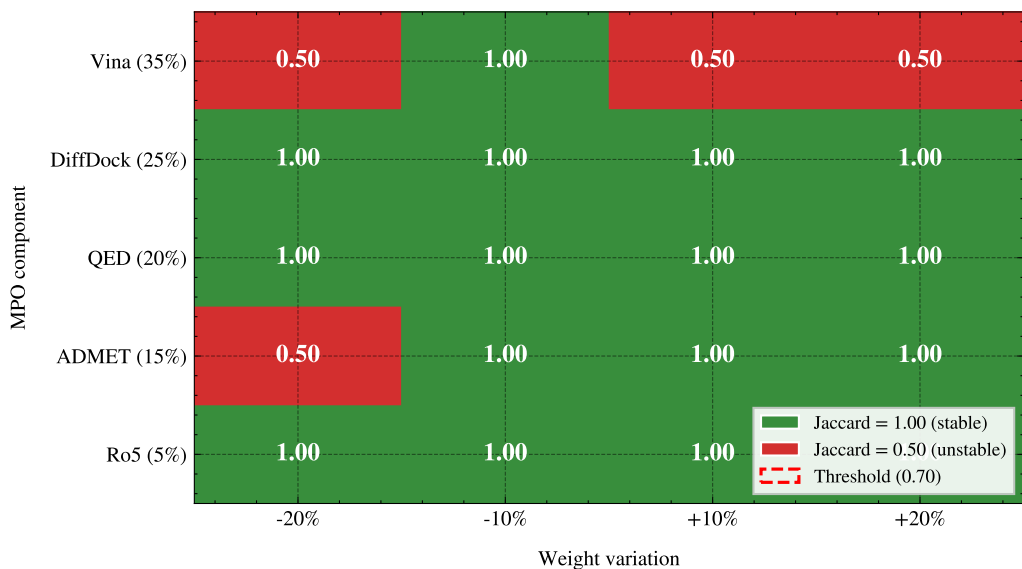


Figure 1: MPO sensitivity analysis heatmap. Each cell shows the Jaccard similarity of the top-20 hit list when the corresponding MPO weight is varied by $\pm 20\%$. Green (1.00) indicates the hit list is unchanged; red (0.50) indicates substantial reordering.

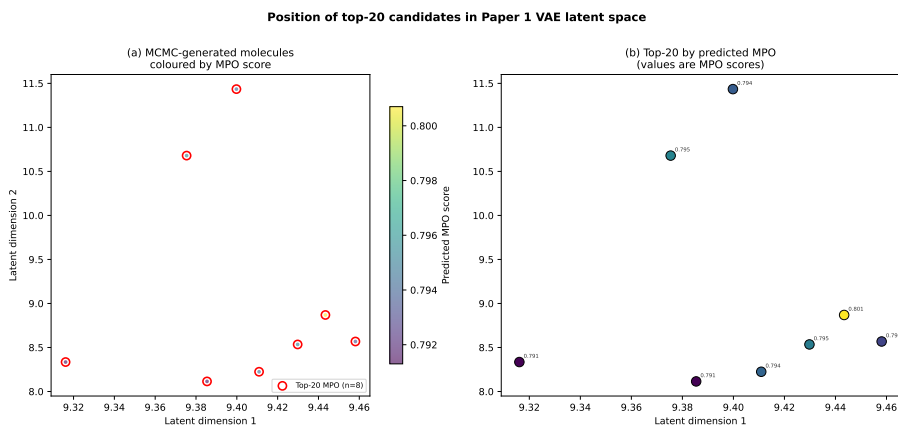


Figure 2: Position of the top-20 candidates in the Paper 1 VAE latent space. Molecules are coloured by predicted MPO score; top-20 by MPO are highlighted in red.

Table 3: ADMET cross-validation for the 20 selected candidates and 5 reference controls. Predictions from three independent tools (ADMET-AI, ADMETlab 3.0, SwissADME) are compared. Spearman correlations between ADMET-AI and ADMETlab 3.0 are reported for solubility (LogS), hERG inhibition (pIC_{50}), blood-brain barrier penetration (logBB), and CYP3A4 inhibition (probability). Compounds where any two tools disagreed by more than one risk category are flagged.

Compound	ADMET-AI LogS	ADMETlab LogS	SwissADME LogS	hERG pIC_{50}	Flagged?
<i>Data to be populated from three independent ADMET tools.</i>					

Table 4: ACSI weight sensitivity analysis. Each of the four ACSI weights (w_{DrugBank} , w_{ANPDB} , w_{fsp^3} , w_{NPL}) was varied by $\pm 20\%$ while keeping the remaining weights proportional. The Jaccard index measures overlap of the resulting top-20 ACSI-ranked list with the baseline ranking. A Jaccard ≥ 0.70 (14/20 compounds stable) indicates robustness.

Weight varied	Variation (%)	Mean ACSI (top-20)	Std.dev.	Jaccard vs. baseline
<i>Data to be populated after ACSI computation.</i>				



Figure 3: Per-compound RRS profiles across all six resistance mutations, shown as radar plots for the 14 polypharmacological leads. Each axis represents one mutation; concentric rings indicate 50 %, 100 %, and 150 % RRS (WT-normalised). Class A* and Class A compounds fill the outer region (RRS > 100 %).